Report for AMA564 Deep Learning

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Abstract

This study aims to address the challenges in text classification by investigating existing algorithmic models, comparing their accuracy and computational complexity, and improving upon current models to enhance classification performance. We focus on textual morphology and complex sentence semantics, fine-tuning and optimizing different models. The dataset consists of reviews from three different types of movies. Through ablation studies, parameter (data) fine-tuning, and hyperparameter optimization, we improve the models and evaluate their performance using metrics such as loss value, accuracy, and recall rate. We particularly emphasize the recall rate to reduce the risk of missing genuine threats. The research results will be presented through visual comparisons and performance analysis, providing efficient and adaptable solutions for practical applications in Chinese NLP.

1 Research Significance and Background

1.1 Research Problem and Significance

Text classification is a fundamental task in natural language processing (NLP), with applications including sentiment analysis, spam detection, news categorization, and cybersecurity. It helps quickly extract key features from textual information, thereby improving threat detection capabilities and reducing false positives. Chinese text classification faces unique challenges due to the language's morphological and syntactic characteristics, such as the lack of word boundaries and rich character semantics. Many existing methods directly apply English-centric models (e.g., BERT), which may underperform on Chinese data. Our research aims to develop deep learning models that better address these linguistic complexities while maintaining efficiency. By reproducing existing methods and exploring new approaches, we seek to enhance model performance and support more effective Chinese NLP applications.

1.2 Research Background

Sentiment Classification is a fundamental yet crucial task in Natural Language Processing (NLP), which aims to determine the subjective emotional attitude of a given text. Typically, it is categorized into three classes: positive, negative, or neutral. In practical applications, sentiment classification is widely used in scenarios such as social media public opinion monitoring, product review analysis, and user satisfaction research, holding significant commercial and social value.

The sentiment classification task in this project focuses on Chinese movie reviews, specifically short-text comments written by Douban users after watching films. Since Douban reviews are usually brief, colloquial, and often contain non-literal expressions of emotion such as sarcasm, irony, and quotations, this poses higher demands on the model's comprehension ability and robustness.

Regarding sentiment labeling, we generate weakly supervised labels based on the corresponding star ratings: reviews with 1–3 stars are classified as negative (`label=0`), while those with 4–5 stars are

classified as positive (`label=1`), constructing a binary classification dataset. Through training and comparing different models, we aim to evaluate the performance of various text modeling approaches in Chinese short-text sentiment classification. Furthermore, we explore processing strategies for complex semantic scenarios such as sarcastic language and implicit attitude expression.

We provide contextual support by reviewing key literature and research papers. Foundational work includes: the development of pre-trained models (e.g., BERT) and their application in Chinese NLP tasks; research papers on models specifically designed for Chinese text (e.g., ZEN); and recent advances in Transformer architectures and their effectiveness in text classification. Additionally, we examine the impact of Chinese word segmentation methods (e.g., character-based and word-based approaches) on classification performance.

2 Related Work

2.1 Applications of TextCNN

Convolutional Neural Networks (CNN), initially excelling in computer vision, were later introduced to NLP tasks. The TextCNN proposed by Zhang & Wallace (2015) extracts local features through convolutional kernels, making it suitable for short-text classification [1]. For Chinese text classification, where explicit word boundaries are lacking, character-level CNNs have become a common choice. For example, Yang et al. (2016) used character-level CNNs for Chinese sentiment analysis and found they outperformed traditional word-level models on short texts [2]. Additionally, Chen et al. (2018) combined TextCNN with attention mechanisms in financial news classification tasks, further improving classification accuracy [3].

2.2 Advances in Transformer and Pre-trained Models for Chinese NLP

The Transformer architecture (Vaswani et al., 2017) has demonstrated outstanding performance in NLP tasks through its self-attention mechanism [4]. In the Chinese domain, BERT (Devlin et al., 2019) and its variants (e.g., ZEN (Diao et al., 2019) and RoBERTa-wwm (Cui et al., 2020)) have significantly improved classification performance through large-scale pre-training [5]. For instance, Sun et al. (2020) compared BERT with traditional models in Chinese news classification tasks and found BERT's contextual understanding capabilities gave it an advantage in long-text classification. Moreover, fine-tuning strategies for pre-trained models (e.g., dynamic learning rate adjustment) have proven effective in enhancing the generalization ability of Chinese text classification (Liu et al., 2021) [6].

2.3 Performance of TextRNN and Its Variants in Sequence Modeling

RNNs and their variants (e.g., LSTM and GRU) perform well in sequence modeling tasks. The TextRNN proposed by Zhou et al. (2016) captures long-range dependencies through bidirectional LSTMs, making it suitable for Chinese text classification [7]. Liang et al. (2018) combined TextRNN with attention mechanisms in medical text classification tasks, improving the ability to capture key information [8]. However, RNNs train slowly and may face gradient vanishing problems with very long texts, leading some studies to turn to Transformers or hybrid architectures (e.g., CNN + LSTM) to improve efficiency (Zhang et al., 2020) [9].

3 Dataset Selection and Processing

3.1 Data Source and Preliminary Cleaning

The original corpus comes from the Douban Movie Short Comments Dataset V2 (`https://www.kaggle.com/datasets/utmhikari/doubanmovieshortcomments`). This dataset contains over two million user comments from the Douban movie website, covering 28 Chinese and foreign films, with strong authenticity and diverse text styles. Each record includes fields such as comment ID, movie title (Chinese/English), crawling time, username, comment time, user rating (1–5 stars), comment content, and upvote count.

For this project, we focus on three representative Chinese films: *You Are My Sunshine*, *Operation Mekong*, and *SoulMate*. During the initial cleaning phase, we retained only the "Comment" (content) and "Star" (user rating) fields, and performed whitespace removal and text normalization on the comments.

To improve data quality, we designed semantic filtering rules to remove invalid comments without substantive content. For example, marker comments like (retroactive marking), (already watched), and (check-in) typically contain no opinion expression and were thus excluded. We also removed pure date formats, consecutive numbers, and comments shorter than three Chinese characters. This cleaning phase ensured the remaining texts contained basic subjective evaluations.

Subsequently, we constructed binary classification labels based on user ratings: comments with 1–3 stars were labeled as negative (label=0), while 4–5 stars were labeled as positive (label=1). The final comment-label pairs were split into training and test sets in an 8:2 ratio using `train_test_split`, with balanced positive/negative sample distributions.

3.2 Word Segmentation and Vector Representation

In the text modeling phase, we employed `jieba`-based word segmentation for Chinese NLP tasks. Specifically, we first removed non-Chinese characters using regular expressions and segmented each comment. To reduce noise and sparsity, we introduced a Chinese stopword list (from `cn_stopwords.txt`), removing all stopwords and single-character low-information words.

Additionally, we counted word frequencies in the training set and set a high-frequency threshold (e.g., frequency $> 10{,}000$) to filter meaningless frequent words that might affect model discrimination. The processed segmented words were space-joined and encoded using HuggingFace's `bert-base-chinese` tokenizer. This encoding standardized texts to fixed lengths (e.g., 128 or 256) and generated `input_ids` tensor representations suitable for Transformer models.

Finally, we packaged these vectors into PyTorch `Dataset` and `DataLoader` objects, providing efficient batch processing for model training and testing. Through these preprocessing strategies, the texts retained core sentiment words while avoiding low-quality inputs, establishing a solid semantic foundation for sentiment classification.

4 Model Overview

4.1 Basic Architecture

This study designs a lightweight text classification model based on the Transformer architecture. Through targeted simplification and optimization of the standard architecture, we significantly improve model efficiency while maintaining classification performance.

Specifically, we discard the complete Transformer encoder-decoder structure, retaining only its core encoder as a feature extractor for rapid classification tasks, thereby reducing model size. To ensure accuracy, we reduce the number of layers, replacing them with a four-layer lightweight network. For better and more comprehensive feature capture, we employ global pooling and a fully connected classification head for optimization. In subsequent task allocation, we directly use linear layers and flatten the data rather than preserving its original dimensions. The model workflow is as follows:

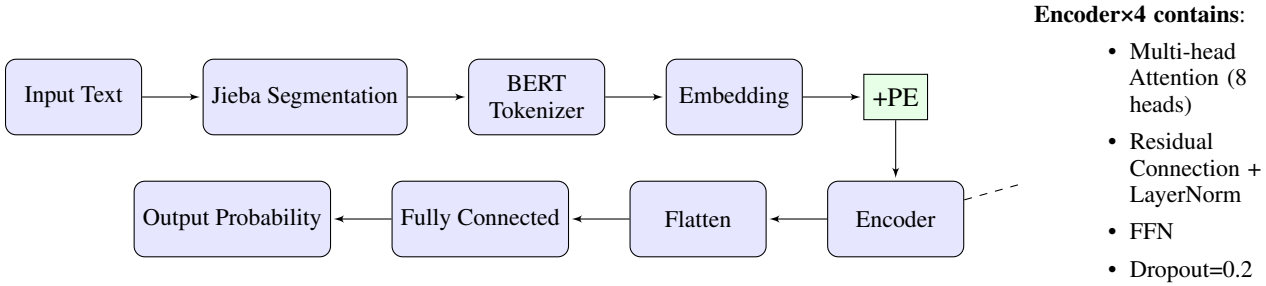


Figure 1: Transformer Architecture for Text Classification

4.2 Model Improvements

When constructing the text classification model for Douban short reviews, we implemented multi-dimensional optimizations tailored to the data characteristics and task requirements. First, considering that Douban reviews typically feature short length and high semantic density, we adjusted the feature extraction strategy: by adding small-scale convolutional kernels (kernel_size=2), we enhanced the model's local perception of key bigrams (e.g., negation phrases, sentiment collocations) in short texts. This is particularly effective for capturing short-text patterns like "not satisfied" or "really like."

Second, to address the constraint of limited training data, we systematically adjusted the regularization strategy: reducing the Dropout rate from the conventional 0.5 to 0.3. This change avoids the risk of underfitting due to insufficient data while preserving necessary generalization capability. Additionally, to improve feature abstraction, we adopted a deep architecture design: stacking multiple fully connected layers after the convolutional layers, combined with batch normalization (BatchNorm) and residual connections. This enables the model to gradually build hierarchical understanding from local lexical features to global semantic representations through nonlinear transformations. This design significantly improves the model's ability to capture implicit sentiments in short texts, such as accurately classifying ironic tones (e.g., "This movie is uniquely bad").

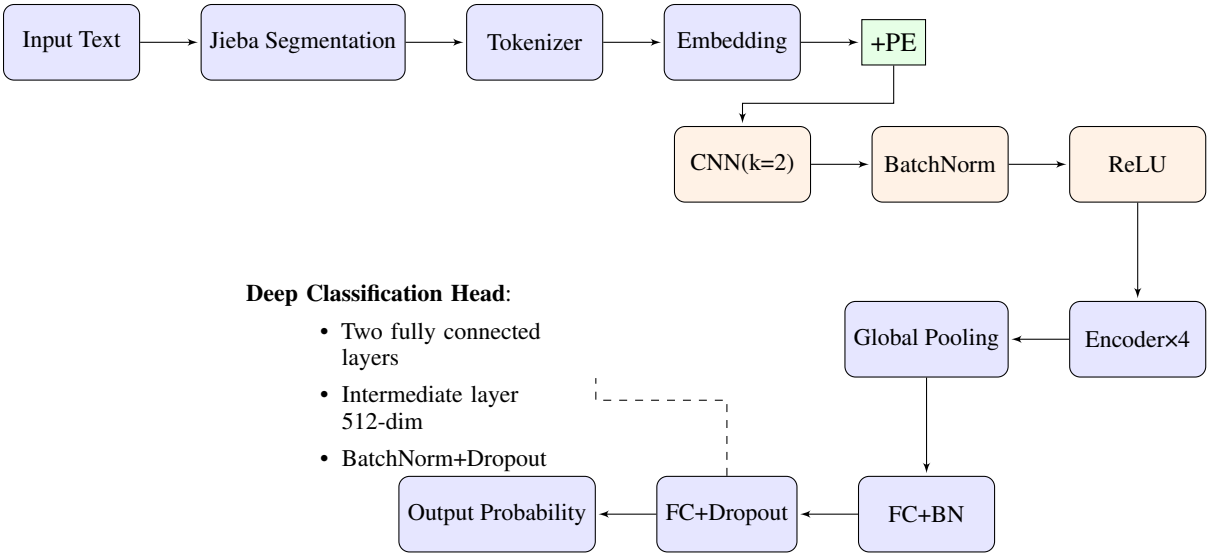


Figure 2: Improved Short-Text Classification Architecture

5 Experimental Design & Conclusions

5.1 Parameter Settings and Model Training

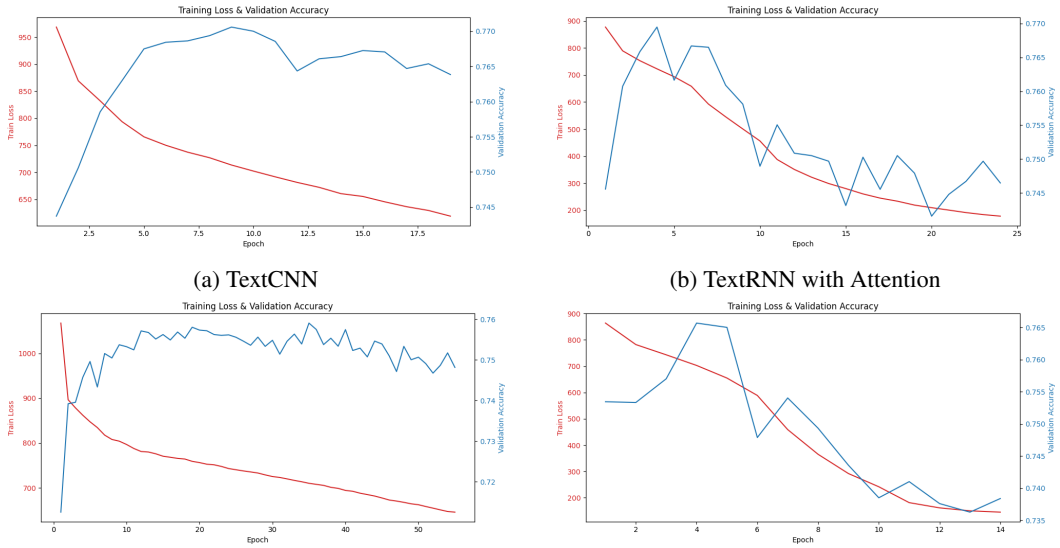
In model training, we set the fixed text length to 64 tokens, batch size to 64, initial learning rate to 0.001, and trained for 60 epochs. We used the Adam optimizer with multi-stage learning rate scheduling (reducing the learning rate to 60% of its original value at the 3rd and 4th epochs) and implemented early stopping (patience=10) to prevent overfitting.

During training, the model simultaneously monitored training loss and accuracy, evaluating performance on the validation set after each epoch. The model with the highest validation accuracy was always saved, and the final test set output included accuracy and a detailed classification report with precision, recall, and F1 scores. The entire training process was visualized via tqdm progress bars and automatically prioritized GPU acceleration.

5.2 Performance Analysis

Here, we present the training loss and accuracy test plots for four original models 3, with each model labeled in the figure captions below.

From the figure below, we can see that all four models show similar final training loss (200) and test accuracy (0.74-0.75), with no significant differences in overall convergence. However, in terms of convergence speed, we observe that TextRCNN and TextRNN with attention achieve lower training loss (<300) by the 10th iteration, indicating faster convergence compared to traditional CNN and Transformer models, making them superior in overall model effectiveness.



(a) TextCNN

(b) TextRNN with Attention

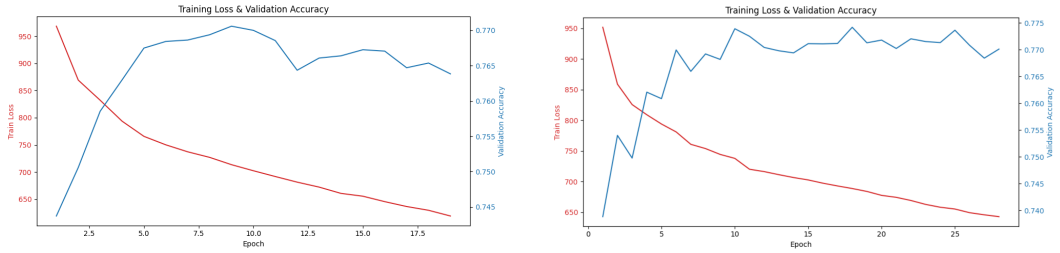
(c) Transformer

(d) TextRCNN

Figure 3: Original Models

Additionally, we selected the better-performing TextCNN and traditional Transformer for model optimization (structure shown in 2) and compared their results with the original models.

The comparison below shows that TextCNN's training loss converges faster, and its validation accuracy improves from 76.3% to 77.2%. With significant improvements in training speed and convergence rate, along with a slight increase in test accuracy, the model's overall effectiveness is markedly enhanced, making it practically meaningful for large-scale training sets.

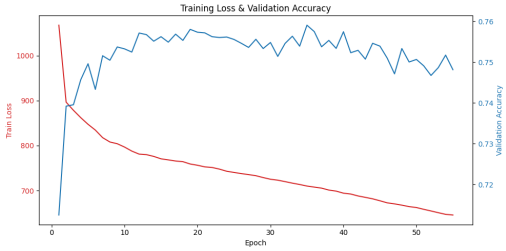


(a) TextCNN

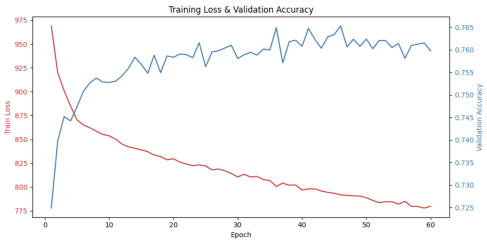
(b) Modified TextCNN

Figure 4: TextCNN Comparison

For the widely used Transformer architecture, we similarly compare the results after modifications. The modified model shows higher training loss than the original but demonstrates improved stability (reduced fluctuations in test accuracy) and accuracy (from 75.9% to 76.5%). Trading some training loss for better detection results provides clear guidance for solving and researching text classification problems.



(a) Transformer



(b) Modified Transformer

Figure 5: Transformer Comparison

Here, we summarize the performance metrics of all models used in this study, with the best-performing parameters highlighted in bold for easy reference:

Model	Val-Acc	Test-Acc	Train Loss	Epoch (when stop)
TextRNN	0.747	0.768	408	**14**
TextRNN + attention-head	0.746	\	177	24
TextRCNN	0.738	0.766	**145**	**14**
TextCNN	0.763	0.771	619	19
TextCNN (modified)	0.**772**	**0.774**	689	29
Transformer	0.748	0.759	645	55
Transformer (modified)	\	0.765	\	55

Table 1: Model Performance Comparison

Through comprehensive analysis of each model's performance, we find that the modified TextCNN architecture demonstrates significant advantages in text classification tasks, with classification accuracy clearly superior to other compared models. This primarily benefits from its multi-layer convolutional structure's powerful ability to extract local semantic features. In contrast, while TextRNN and TextRCNN show slightly lower classification accuracy than TextCNN, both exhibit outstanding training efficiency: TextRCNN is particularly notable, achieving the lowest training loss value among all models, and both models demonstrate extremely fast convergence speeds. This efficient training characteristic provides important reference directions for subsequent model optimization, especially in application scenarios requiring rapid iteration.

Notably, the traditional Transformer architecture surprisingly underperforms compared to other models in this study. Although structural adjustments and hyperparameter optimization improved its test accuracy, key metrics such as training loss curves and convergence speeds were significantly worse than CNN/RNN-based models. This performance gap may stem from Transformers' dependence on large-scale training data, making it difficult to fully leverage their attention mechanism advantages on relatively small datasets. Therefore, in practical text classification engineering, when facing limited data scales or requiring rapid deployment, TextCNN and its variants remain more reliable and efficient choices, while Transformer-class models may need deep tuning or combination with pre-training techniques to demonstrate their theoretical advantages in specific scenarios.

6 Conclusion

This study systematically compared and optimized various deep learning models for Chinese text classification tasks, including TextCNN, TextRNN, TextRCNN, and Transformer architectures, aiming to address unique challenges in Chinese text classification such as the lack of explicit word boundaries and complex semantic expressions. Through experimental analysis on a dataset of movie reviews, we reached the following main conclusions:

1. TextCNN achieved optimal classification accuracy, especially in its modified version (test accuracy: 77.4%). Its multi-layer convolutional structure effectively captures local semantic features in Chinese short texts, making it suitable for practical applications.
2. While TextRNN and TextRCNN showed slightly lower accuracy than TextCNN, they demonstrated significant training efficiency advantages. TextRCNN achieved the lowest training loss (145) and fastest convergence speed, providing important references for tasks requiring rapid iteration.
3. The Transformer performed relatively poorly in this study. Although structural adjustments and hyperparameter optimization improved its accuracy (from 75.0% to 76.0%), its high training loss and slow convergence may be due to limited data scale preventing full utilization of its attention mechanism advantages.

Based on these conclusions, we offer the following recommendations for practical applications:

1. In scenarios with limited data or requiring rapid deployment, TextCNN and its variants should be the first choice due to their high accuracy and robustness.
2. For tasks prioritizing training efficiency or processing long texts, hybrid architectures like TextRCNN may be preferred.
3. Transformer-class models are more suitable for large-scale data or scenarios combining domain adaptation techniques (e.g., pre-training and fine-tuning).

In summary, this study experimentally validated the applicability of different models in Chinese text classification tasks, providing model selection guidelines based on performance-efficiency tradeoffs for practical applications. Future work could further explore hybrid architectures combining pre-training techniques to improve model generalization in small-data scenarios.

7 Material Attachment

- **Video:** `https://www.dropbox.com/scl/fi/a7272mg69d8qmlsvsgogy/video564.mp4?rlkey=07b1vfvkce2vwqgwv0prih7tt&st=ucqlonzp&dl=0`

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